

StyCLIP: StyleTransfer in Few-Shot Learning

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Introduction & Motivation

- **Challenge:** Visual artist classification is crucial for art history and content analysis, especially with growing digital collections.
- **Focus:** Few-shot learning (FSL) for artist classification – identifying artists with limited labeled examples.
- **Problem:**
 - Artistic style involves subtle visual cues (brushstrokes, color palettes, composition).
 - CLIP-based FSL methods excel at semantics but might overlook fine-grained stylistic nuances.
- **Hypothesis:** Integrating explicit style representations (Gram matrices) into CLIP could enhance artist-specific pattern capture.
- **Proposal:** StyCLIP, a CLIP extension incorporating style features.

Style Transfer

- Gatys et al. [3]: Deep networks encode style in intermediate layers; Gram matrices capture style.

Few-Shot Learning (FSL) & Meta-Learning

- Prototypical Networks [6]: Learn to learn from small samples.

Contrastive Language-Image Pretraining (CLIP)

- CLIP [5]: Shared embedding space for images and text.
- TIP-Adapter [8]: Non-parametric cache model for FSL.
- APE [9]: Adaptive prior refinement of CLIP features.
- GDA-CLIP [7]: Gradient-based adaptation.
- TIMO [4]: Mutual guidance between text/image modalities.

Background: TIMO (Text-Image Mutual Guidance Optimization)

TIMO (Text-Image Mutual Guidance Optimization) improves training-free Few-Shot Learning with CLIP by addressing:

Two Main Branches

Text-Guided Image (TGI):

- Uses textual representations to guide visual features
- Computes similarity weights between prompts and visual prototypes
- Enriches visual features with weighted textual information

Image-Guided Text (IGT):

- Improves textual prompt quality using visual guidance
- Finds optimal weights for textual prompts
- Creates rectified textual prototypes

Final Combination

$$\text{logits}_{\text{TIMO}} = \text{logits}_{\text{TGI}} + \alpha \cdot \text{logits}_{\text{IGT}}$$

Dataset: ArtGraph Subset

- Derived from ArtGraph dataset [2] ($>100,000$ paintings).
- Two non-overlapping subsets created:

ArtGraph Few-Shot Subset (Evaluation)

- **Style Selection:** Top 10 styles by artwork count.
- **Artist Filtering:** 30–50 artworks/style (up to 10 artists/style).
- **Splits:** 80% support, 10% validation, 10% query.

ArtGraph StyCLIP Subset (Training)

- Complementary artists (16–50 artworks).
- Excluded from few-shot subset (no data leakage).

StyCLIP: Architecture Overview

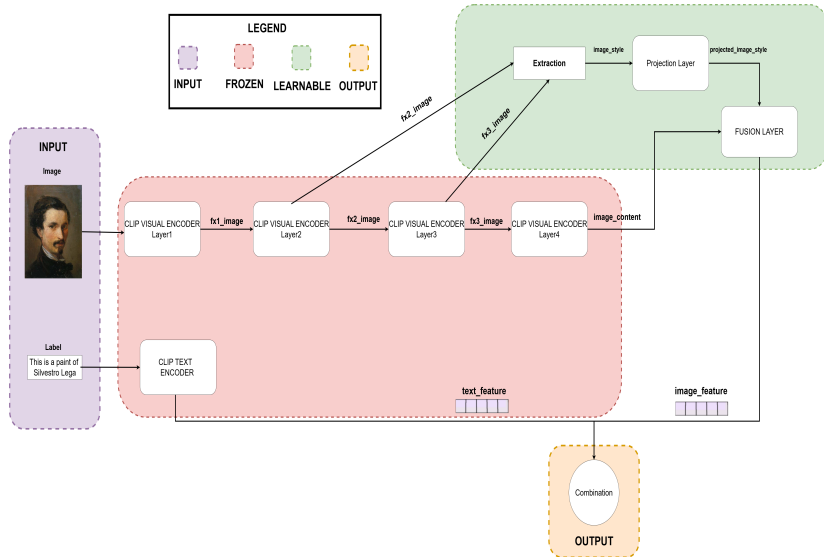


Figure: StyCLIP Architecture Overview

StyCLIP: Key Components

- **Core Idea:** Extend CLIP with explicit style-aware features via Gram matrices
- **Three Key Components:**
 - ① **Gram Matrix Extraction:** Capture style correlations from intermediate visual layers (ResNet-50 blocks)
 - ② **Style Feature Projection:** Reduce dimensionality of Gram matrix representations
 - ③ **Fusion Adapter:** Integrate semantic (CLIP) and style features through bottleneck architecture

StyCLIP: Training Strategy

- **Frozen Backbone:** Original CLIP (RN50) visual and text encoders are frozen.
- **Trainable Components:** Only the new style projection layers and fusion adapter are optimized.
- **Rationale:**
 - Preserve pre-trained semantic knowledge from CLIP.
 - Reduce computational cost.
 - Enable efficient fine-tuning on artistic datasets.

Hyperparameter Tuning

- Optuna [1] used for:
 - Fusion bottleneck dimension $\{64, 128, 256\}$
 - Dropout rate $[0.0, 0.5]$
 - Learning rate log-uniform $[10^{-5}, 10^{-3}]$
 - Weight decay log-uniform $[10^{-6}, 10^{-3}]$

Experimental Setup

Meta-Learning Training for StyCLIP

- Prototypical network approach.
- Episodic training: N-way ($N=10$ artists) classification.
- K support examples, $Q=3$ query examples per artist per episode.
- Two StyCLIP training configurations:
 - **StyCLIP FLS1**: Trained with $K=1$ support shot.
 - **StyCLIP FLS4**: Trained with $K=4$ support shots.

Evaluation Protocol

- N-way classification (all available artists in the test subset).
- Tested across $K \in \{1, 2, 4, 8, 16\}$ shots for support set.
- **Statistical Robustness**: 10 different random seeds.
- **Metrics**: Accuracy, Macro-Precision, Macro-Recall, Macro-F1.

Baseline Methods & Backbones

Few-Shot Adaptation Methods

- CLIP + TIP-Adapter [8]
- CLIP + APE [9]
- CLIP + GDA-CLIP [7]
- CLIP + TIMO [4]
- CLIP + TIMO-S

Backbones Compared

- **CLIP RN50**: Standard CLIP ResNet-50.
- **StyCLIP FLS1**: Our custom backbone, trained with 1-shot.
- **StyCLIP FLS4**: Our custom backbone, trained with 4-shots.

An example episode for RN50:

Hyperparameter Optimization Highlights

StyCLIP FLS1 (1-shot trained)

- Bottleneck: 64
- LR: 6.01×10^{-5}
- Dropout: 0.0
- Weight Decay: 4.83×10^{-4}
- Valid Acc: 57.07%

StyCLIP FLS4 (4-shot trained)

- Bottleneck: 128
- LR: 1.02×10^{-5}
- Dropout: 0.1
- Weight Decay: 1.03×10^{-6}
- Valid Acc: 60.53%

Example Episode for RN50



Figure: Example episode: 5-Way, 4-Shots ,10 query examples

Key Evaluation Results

Main Finding: Integrating explicit style-aware features (StyCLIP) *consistently underperforms* standard CLIP variants.

- Standard CLIP with RN50 backbone is the best performing backbone.
- StyCLIP backbones (Custom_FSL1_RN50, Custom_FSL4_RN50) significantly degrade performance for all FSL methods.
- TIMO and TIMO-S (our adaptation logic) are the best FSL methods when using the standard RN50 backbone.

Example: 16-Shot Classification Accuracy (%)

Model	Backbone	ACC (%)
TIMO	Custom_FSL1_RN50	37.00
TIMO	Custom_FSL4_RN50	43.27
TIMO	RN50	73.36
TIMO_S	Custom_FSL1_RN50	39.20
TIMO_S	Custom_FSL4_RN50	48.75
TIMO_S	RN50	73.46

Statistical Analysis Summary

- Paired t-tests confirmed the significance of performance differences.
- **Key Observation:** Standard CLIP (RN50) significantly outperforms StyCLIP-trained backbones across nearly all configurations and few-shot methods.
- Example: For 16-shot with TIMO_S:
 - RN50 (73.46%) vs Custom_FSL4_RN50 (48.75%) $\rightarrow \approx 25$ percentage point difference, statistically significant.

$\Rightarrow$ The architectural changes in StyCLIP for explicit style modeling did not yield benefits and often hindered performance.

Conclusion & Discussion

- **Main Finding:** Integrating explicit style-aware features via Gram matrices (StyCLIP) did **not improve** few-shot artist classification performance.
- **Performance:** Standard CLIP (RN50 backbone) consistently outperformed StyCLIP variants.
- **Implication:** Explicit Gram matrix-based style modeling might interfere with CLIP's pre-trained semantic representations rather than enhance them for this task.
- **Future Directions:** Alternative approaches like attention-based style integration or domain-specific fine-tuning (without architectural changes) might be more fruitful.

Acknowledgments

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The code for this project is available on GitHub:

<https://github.com/Fonty02/ComputerVision/tree/main/TIMO>

Thank You!

References I



Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama.

Optuna: A next-generation hyperparameter optimization framework.

In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, pages 2623–2631, 2019.



Giovanna Castellano, Vincenzo Digeno, Giovanni Sansaro, and Gennaro Vessio.

Artgraph, March 2022.



Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge.

Image style transfer using convolutional neural networks.

In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.



Yayuan Li, Jintao Guo, Lei Qi, Wenbin Li, and Yinghuan Shi.

Text and image are mutually beneficial: Enhancing training-free few-shot classification with clip.

In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 5039–5047, 2025.



Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al.

Learning transferable visual models from natural language supervision.

In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.



Jake Snell, Kevin Swersky, and Richard Zemel.

Prototypical networks for few-shot learning.

In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.



Zhengbo Wang, Jian Liang, Lijun Sheng, Ran He, Zilei Wang, and Tieniu Tan.

A hard-to-beat baseline for training-free clip-based adaptation.

arXiv preprint arXiv:2402.04087, 2024.

References II



Renrui Zhang, Wei Zhang, Rongyao Fang, Peng Gao, Kunchang Li, Jifeng Dai, Yu Qiao, and Hongsheng Li.

Tip-adapter: Training-free adaption of clip for few-shot classification.

In *European conference on computer vision*, pages 493–510. Springer, 2022.



Xiangyang Zhu, Renrui Zhang, Bowei He, Aojun Zhou, Dong Wang, Bin Zhao, and Peng Gao.

Not all features matter: Enhancing few-shot clip with adaptive prior refinement.

In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2605–2615, 2023.